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The densities of liquid argon, krypton xenon, oxygen, nitrogen, carbon monoxide methane, and carbon tetrafluoride along the orthobaric liquid curve

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The densities along the orthobaric liquid curve of the following liquefied gases have been measured with a precision of a few parts in 10^4: argon (86 to 118 K); krypton (118 to 164 K); xenon (164 to 219 K); oxygen (80 to 121 K); nitrogen (78 to 105 K); carbon monoxide (78 to 111 K); methane (92 to 151 K); carbon tetrafluoride (91 to 185 K). The measurements were made on a sample of about 9 cm^3 in a glass pyknometer, which was incorporated in a cryostat assembly similar to that used in conventional low-temperature adiabatic calorimetry.

The results have been considered from the point of view of the law of corresponding states. When the volume is plotted as a fraction of the critical volume V_c against temperature expressed as a fraction of the critical temperature T_c, the results for liquid krypton, xenon, and nitrogen conform to a common curve. By changing the "best" values of V_c for argon and oxygen by about 0.5 per cent (which is probably within the uncertainty limits of these values of V_c), the points for these two liquids could also be made to fall on the same curve. But there are indications of small divergences from this curve for methane and carbon monoxide, even if the values of V_c adopted for these two substances are altered.

Over the temperature ranges studied, the so-called rectilinear diameter for all eight substances examined is in fact a very slight curve, but extrapolation to the critical temperature of a least-squares straight line gives values of the critical volume within about one per cent of the literature values.

1. Introduction

Several experimental determinations of the excess thermodynamic functions of binary mixtures of condensed gases have been made in this laboratory, primarily to obtain results to test modern theories of solutions. The application of some of these theories to individual systems requires reliable information about certain physical properties of the pure components. One of these properties is the density of the liquid as a function of temperature. Information on the density of even quite common liquefied gases is often fragmentary and not particularly accurate. We have therefore thought it worthwhile to measure the densities of the pure liquefied gases involved in our studies of binary systems. The measurements were made at the

saturation vapour pressure, i.e. along the orthobaric liquid curve. They usually started a kelvin or two above the triple-point and were continued until the vapour pressure reached about 10 atm, this being more or less the practical upper limit with the apparatus used. It is hoped that the absolute error in our results does not exceed about 0.1 per cent. The same apparatus was used for all the gases studied, which is clearly an advantage in situations where comparison is important. After presenting our results, we shall briefly consider their application to two matters of some general interest, namely the representation of liquid volumes on a corresponding states basis, and the so-called law of the rectilinear diameter.

2. Experimental

The density determinations were made with a glass pyknometer of known volume (about 9 cm^3). The amount of gas needed to fill this with liquid when it was maintained at the lowest temperature of a run was first determined, and then the temperature of the pyknometer was raised in stages of 2 to 5 K, and the amount of gas evaporated off in each stage measured. The pyknometer was incorporated in a cryostat very similar to an adiabatic calorimeter, with the place of the sample container of the calorimeter taken by a metal block with a cavity for the pyknometer. The necessary visibility of the capillary inlet tube inside the metal cryostat was secured by a periscope system of mirrors and a small internal light. While this work was in progress a description of a similar apparatus (used to determine the density of liquefied carbon tetrafluoride) was published by Knobler and Pings.[1]

Some of the details of our apparatus are shown in figure 1. The pyknometer was of Pyrex with a wall thickness of approximately 4 mm. Its volume at 0 °C was determined by water-filling, and that at lower temperatures estimated by using the coefficients of expansion of Buffington and Latimer.[2] The increase in volume with internal pressure for pressures up to 20 atm was found by direct experiment to be 5.8×10^{-5} cm^3 atm^{-1} at 0 °C, and this coefficient was assumed to be temperature-independent. The graduated part of the capillary inlet tube to the pyknometer was of about 1.5 mm i.d. for the first two cm or so, and about 1 mm i.d. for the upper two cm. At the lower temperatures of a run, the narrower part of the tube could be used to obtain a more accurate volume measurement. At higher temperatures, it was more convenient to use the lower, wider tube. Since the capacity of the graduated part of the capillary tube was relatively little, small safety bulbs were blown above and below the graduated capillary, as shown in figure 1, so that measurements were made with liquid in the lower bulb and gas in the upper one. The emergent glass capillary above the upper bulb was narrow (about 0.7 mm i.d.) so as to reduce the amount of gas in the region where there was a temperature gradient. The glass pyknometer was a sliding fit in a cylindrical cavity in an aluminium block, which was also drilled to take a platinum resistance thermometer and provided with a heating coil on its surface. The block was surrounded by a copper radiation shield. It was of course essential that there should be no point on the inlet tube to the pyknometer colder than the pyknometer itself, but that at the same time the temperature of the liquefied gas in the small safety bulb and the graduated capillary should be as near

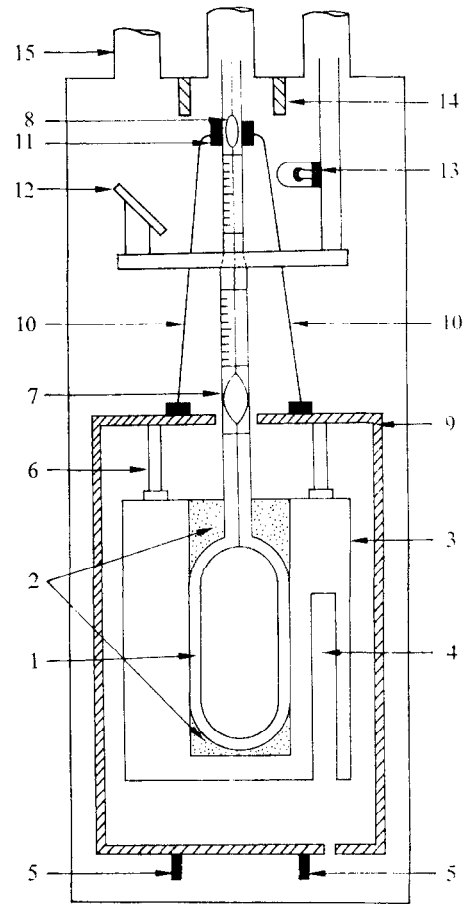


FIGURE 1. Pyknometer and cryostat assembly. 1, Glass pyknometer; 2, paraffin wax and copper filings; 3, aluminium block; 4, cavity for platinum resistance thermometer; 5, distribution panels for electrical leads; 6, plastic supporting rods; 7, 8, safety bulbs; 9, copper radiation shield; 10, flexible copper strips; 11, copper collar; 12, stainless steel mirror; 13, small light; 14, copper cylinder round which electrical leads are wound; 15, viewing tube.

as possible to that of the pyknometer. These requirements were achieved by use of flexible copper strips, soldered at their lower ends to the roof of the shield, and at their upper ends to a copper collar clamped to the glass tube. Calibrated thermocouples on the block and collar gave the temperature difference between them, which with practice could be kept at 1 to 2 cK when measurements were being made.

The internal mirror of the periscope system (of polished stainless steel) and the small light to illuminate the scale and meniscus were mounted on the same metal ring. This ring was fixed to a rod which passed out of the top of the cryostat through a double O-ring seal, so that the internal light and mirror could be moved up and down over about 6 cm as required. The light had an appreciable heating effect (which

was sometimes useful), and when readings were taken it was switched on for the shortest possible time. A small frosted glass plate (not shown in figure 1) was fixed between the light bulb and the glass capillary. The meniscus was observed with a telescope, and its position could be estimated to 0.1 mm corresponding to a volume of about 2×10^{-4} cm^3 in the wide capillary and about 0.8×10^{-4} cm^3 in the narrower.

The amount of gas needed to fill the pyknometer was determined by pressure measurements made with a constant volume mercury manometer, constructed of precision bore tubing of 14 mm i.d. The gas was contained in calibrated globes of volume about 8 dm^3 submerged in a thermostat at a temperature near 25 °C and held constant to ± 0.01 K, and its pressure was measured before and after condensation into the pyknometer. Depending on the vapour pressure of the liquefied gas at the starting temperature of a run, the gas was either condensed directly into the pyknometer, or first into a trap cooled with liquid nitrogen and from there evaporated into the pyknometer. The aluminium block in the cryostat was held at a temperature constant to a few mK, and the position of the liquid meniscus on the scale measured. The temperature of the block was then raised by from 2 to 5 K, and gas released from the sample through a metal fine-control valve to maintain the meniscus at a suitable position on the scale. Usually the gas was evaporated directly into a calibrated vessel of capacity about 224 cm^3 which was also in the 25 °C thermostat and in connexion with the constant-volume manometer. But when the vapour pressure was small, the gas so released was first condensed in liquid nitrogen and evaporated from there into the calibrated vessel. Near the triple-point, the evaporation, if carried out too quickly, produced surface freezing in the capillary. Since the capillary had been accurately calibrated, it was unnecessary to try and achieve the same reading of the meniscus after each change of temperature.

The volumes of dead-spaces were made as small as possible, and were measured by the usual gas-displacement technique. In assessing the amount of substance needed to fill the pyknometer, corrections were applied for gaseous imperfection and for the small change in the internal volume of the calibrated globes with pressure. At the end of a run, the liquid in the calibrated pyknometer was evaporated into the globes and a check made of the sum of the amount of this gas plus the amounts evaporated off during the run, as compared with that originally condensed into the pyknometer.

The platinum thermometer was of the coiled-coil type in a platinum capsule. There was no reason to distrust its reliability, but a check measurement was made of the normal boiling temperature of methane, which gave 111.664 K, in good agreement with Lovejoy's[3] value of 111.657 K.†

The gases used were the purest commercially available, each gas being further purified by fractionation in the column described by Davies *et al.*[4] The purity of the sample on which the density measurements were made was kindly checked for us with a mass-spectrometer by Dr. C. J. Danby. In no case were there any signs of peaks due to impurities; this indicates a purity of at least 99.99 moles per cent. In addition, for three gases—methane, carbon monoxide, and xenon—measurements were made of the triple-point pressures, all of which were constant during melting

† Throughout this paper temperatures, denoted by T, were measured on the International Practical scale of 1948, and K is used to denote K(int.1948).

to ± 0.02 mmHg with mean values of 87.64 mmHg for CH_4, 115.31 mmHg for CO, and 611.66 mmHg for Xe.†

3. Results

The experimental results are given in table 1 as molar volumes. They were smoothed by fitting them to polynomials in $(T-T_r)$, T being the thermodynamic temperature and T_r a convenient reference temperature. For argon, krypton, xenon, methane,

TABLE 1. Experimental values of the liquid molar volume V_m at the saturation vapour pressure. The last column gives the value δ of the experimental molar volume less the value calculated from the equation of table 2

Run	T / K	V_m / cm^3 mol^{-1}	δ / cm^3 mol^{-1}	Run	T / K	V_m / cm^3 mol^{-1}	δ / cm^3 mol^{-1}
			Argon				
1	85.658	28.513	−0.001	1	104.365	31.210	−0.002
2	87.279	28.713	−0.001	2	106.992	31.675	0.005
1	89.919	29.056	−0.003	1	109.408	32.127	0.004
2	92.148	29.353	−0.001	2	111.234	32.474	−0.002
1	94.793	29.725	0.001	2	113.538	32.952	−0.002
2	97.452	30.109	−0.004	1	115.108	33.288	−0.012
1	100.147	30.528	0.003	2	116.433	33.619	0.014
2	102.083	30.831	−0.002	2	117.854	33.944	−0.003
						mean magnitude:	0.004
			Krypton				
2	117.703	34.510	−0.002	2	140.386	37.248	0.002
1	118.121	34.552	−0.003	1	142.778	37.619	−0.004
2	119.770	34.731	0.002	2	144.527	37.884	0.008
1	121.837	34.955	0.000	1	146.816	38.229	0.005
2	123.815	35.185	0.008	2	149.036	38.580	0.010
1	126.061	35.439	0.002	1	152.183	39.082	0.000
2	128.015	35.676	0.006	2	154.348	39.440	−0.011
1	130.304	35.949	−0.014	1	155.990	39.727	−0.013
2	131.840	36.146	0.003	2	157.827	40.083	0.008
1	134.253	36.447	−0.006	1	159.949	40.482	0.005
2	135.969	36.679	0.000	2	162.400	40.962	−0.002
1	138.459	37.010	−0.004	1	163.936	41.284	0.001
						mean magnitude:	0.005
			Xenon				
3	164.084	44.579	0.001	2	194.655	48.071	−0.003
3	167.506	44.925	0.003	3	199.915	48.805	0.015
1	171.512	45.332	−0.013	2	203.453	49.279	−0.014
3	174.849	45.704	0.009	3	206.713	49.785	0.014
2	179.368	46.188	−0.009	2	210.747	50.355	−0.029
3	183.314	46.674	0.020	3	214.744	51.041	0.028
2	186.935	47.083	−0.007	2	219.069	51.709	−0.009
3	191.009	47.594	−0.006				
						mean magnitude:	0.012

† mmHg = (101.325/760) kN m^{-2}.

TABLE 1. *Continued*

Run	T / K	V_m / cm^3 mol^{-1}	δ / cm^3 mol^{-1}	Run	T / K	V_m / cm^3 mol^{-1}	δ / cm^3 mol^{-1}
			Oxygen				
2	80.095	26.980	−0.007	1	99.788	29.396	−0.002
1	80.156	27.000	0.006	2	101.851	29.699	0.003
2	83.210	27.321	0.000	1	104.797	30.140	−0.003
1	84.274	27.438	−0.001	2	105.386	30.238	0.003
2	86.828	27.734	0.003	1	109.445	30.899	−0.002
1	88.046	27.874	0.000	2	110.275	31.052	0.007
2	90.617	28.188	0.003	1	113.973	31.713	0.005
1	91.777	28.326	−0.003	2	114.429	31.811	0.006
2	94.517	28.681	0.001	2	117.581	32.431	−0.004
1	95.340	28.787	−0.001	1	117.943	32.507	−0.004
2	98.149	28.164	−0.004	2	121.399	33.274	0.003
						mean magnitude:	0.003
			Nitrogen				
2	77.594	34.912	0.010	2	92.329	38.350	0.013
1	78.874	35.158	−0.006	1	93.981	38.806	−0.004
2	81.075	35.621	−0.002	2	95.388	39.247	0.015
1	82.467	35.910	−0.013	1	97.184	39.790	−0.008
2	84.504	36.376	0.001	2	100.735	41.001	−0.019
1	86.411	36.825	0.008	2	103.356	42.026	0.008
2	89.129	37.486	0.001	2	104.741	42.585	0.003
1	91.011	37.969	−0.007				
						mean magnitude:	0.008
			Carbon monoxide				
2	77.729	34.775	0.009	1	98.153	39.361	0.025
1	79.995	35.210	0.005	2	99.859	39.821	−0.005
2	82.415	35.666	−0.016	1	101.645	40.389	0.022
1	84.953	36.191	−0.005	2	102.887	40.725	−0.035
2	87.788	36.782	−0.013	1	105.374	41.620	0.026
1	90.067	37.303	0.003	2	106.735	42.025	−0.053
2	92.865	37.956	−0.001	1	108.171	42.626	0.013
1	94.678	38.426	0.018	2	109.578	43.128	−0.030
2	96.399	38.857	0.001	1	111.044	43.791	0.037
						mean magnitude:	0.018
			Methane				
1	92.029	35.795	−0.004	1	123.040	39.773	0.061
2	93.090	35.901	−0.006	2	125.765	40.107	−0.025
1	93.654	35.971	0.006	1	127.426	40.431	0.032
1	95.743	36.197	0.011	2	130.078	40.814	−0.021
2	95.905	36.198	−0.005	1	132.228	41.224	0.019
1	100.939	36.781	0.012	2	136.122	41.901	−0.011
2	102.616	36.961	−0.005	3	137.671	42.153	−0.055
1	107.338	37.552	0.007	2	140.482	42.756	−0.014
2	108.806	37.714	−0.017	3	142.092	43.151	0.043
1	112.156	38.173	0.005	2	145.195	43.792	−0.001
2	114.003	38.394	−0.022	3	147.541	44.369	0.024
1	117.523	38.913	0.009	2	151.369	45.300	−0.016
2	119.319	39.135	−0.025				
						mean magnitude:	0.018

TABLE 1. *Continued*

Run	T / K	V_m / cm^3 mol^{-1}	δ / cm^3 mol^{-1}	Run	T / K	V_m / cm^3 mol^{-1}	δ / cm^3 mol^{-1}
			Carbon tetrafluoride				
1	91.336	47.158	0.019	1	135.142	53.155	−0.015
2	96.188	47.764	0.021	1	139.605	53.892	−0.025
1	96.426	47.753	−0.020	2	139.824	53.950	−0.004
2	98.982	48.054	−0.036	1	144.505	54.764	−0.012
1	100.105	48.218	−0.012	2	145.628	54.977	−0.002
1	103.790	48.678	−0.014	1	150.634	55.901	−0.015
2	103.895	48.705	−0.001	2	150.917	55.975	0.005
1	107.570	49.181	0.006	2	155.296	56.846	0.007
1	112.236	49.851	0.063	1	155.839	56.970	0.020
2	118.237	50.612	0.003	2	160.768	58.033	0.036
1	119.059	50.715	−0.009	1	161.332	58.178	0.056
2	123.293	51.342	0.008	2	165.773	59.140	−0.001
1	124.935	51.570	−0.007	1	167.729	59.564	−0.050
2	128.858	52.179	0.008	2	170.851	60.357	−0.045
1	129.361	52.239	−0.010	2	178.051	62.431	0.020
2	133.921	52.975	0.003	2	184.710	64.568	0.003
						mean magnitude:	0.017

and carbon tetrafluoride, T_r was the triple-point temperature of the substance itself, for carbon monoxide and nitrogen T_r was the normal boiling temperature of nitrogen, while for oxygen T_r was arbitrarily taken as 80 K. The values of the coefficients in the polynomials are given in table 2. An estimate of the precision of the experimental

TABLE 2. Values of the coefficients in the equation:

$$V_m/\text{cm}^3\,\text{mol}^{-1} = V_r/\text{cm}^3\,\text{mol}^{-1} + A(T - T_r)/\text{K} + B(T - T_r)^2/\text{K}^2 + C(T - T_r)^3/\text{K}^3 + D(T - T_r)^4/\text{K}^4 + E(T - T_r)^5/\text{K}^5$$

	T_r/K	V_r/cm^3 mol^{-1}	$10A$	10^3B	10^5C	10^7D	10^9E
Argon	83.82	28.295 8	1.157 71	1.550 95	−3.084 15	8.429 18	—
Krypton	115.8	34.318 9	0.993 967	1.054 18	−1.379 05	2.373 45	—
Xenon	161.38	44.312 0	0.973 706	0.325 239	0.539 56	−0.297 766	—
Oxygen	80.00	26.977 1	1.041 34	0.958 032	−0.811 194	3.118 55	—
Nitrogen	77.355	34.853 5	2.025 00	0.929 986	7.228 96	—	—
Carbon monoxide	77.355	34.693 4	1.929 92	0.160 729	6.214 49	—	—
Methane	90.68	35.666 1	0.973 681	1.127 02	−1.597 88	2.330 39	—
Carbon tetrafluoride	89.569	46.916 6	1.265 79	−0.400 011	2.322 62	−2.718 15	1.474 65

results is given by the values of the mean of the numerical differences between the experimental molar volumes and those calculated from the polynomials (table 1).

Some very accurate density measurements, extending almost to the critical temperature, have recently been made in the laboratories of the British Oxygen Company on liquid argon, nitrogen, and oxygen at the saturation vapour pressure. These

results, obtained by a quite different technique, have been kindly communicated to us before their publication. The agreement between these measurements and ours for oxygen and nitrogen is very gratifying, the average difference between the two sets of results being about 0.05 per cent for each of the two substances, with no sign of any systematic discrepancy. On the other hand, our values of the molar volume of liquid argon are significantly larger than the B.O.C. values, by 0.17 per cent near the triple-point temperature, by 0.11 per cent at 110 K, and by 0.27 per cent at 120 K. Our molar volumes differ by less than 0.1 per cent from the early Leiden results of Mathias, Onnes, and Crommelin.[5] The density of krypton as determined later in the same laboratory by Mathias *et al.*,[6] who seem to have made the only accurate measurements on this element, are in even better agreement with ours, the average difference being 0.03 per cent. Leadbetter and Thomas[7] have recently measured the density of liquid xenon, claiming an accuracy of "a few parts per thousand", and within these limits our results agree with theirs. The most modern determinations on methane are those of Davenport, Rowlinson, and Saville[8] and of Grigor and Steele.[9] The molar volumes recorded by the first group are about 0.15 per cent less than ours, while those of Grigor and Steele are also less than ours, the maximum discrepancy reaching about 0.3 per cent. For carbon tetrafluoride, the recent density results of Knobler and Pings[1] over the range 90 to 150 K correspond to molar volumes consistently about 0.25 per cent below ours. The only measurements on carbon monoxide seem to be those of Mathias and Crommelin,[10] which show a scatter of about 0.1 per cent. Within this scatter, they agree with ours at the highest temperature of our range, about 110 K, but with decreasing temperature their molar volumes fall increasingly below ours, the discrepancy reaching 0.8 per cent at about 80 K.

4. Discussion

Whether or not the molar volumes of the substances studied conform to the law of corresponding states can be examined by plotting V/V_c (or V_c/V) against T/T_c, where V_c and T_c are respectively the critical volume and temperature. Small departures from a common curve are more easily detected by plotting the differences between the experimental values and those given by a suitable reference curve. For the latter we have chosen Guggenheim's equation:[11]

$$V_c/V = 1+\tfrac{3}{4}(1-T/T_c)+\tfrac{7}{4}(1-T/T_c)^{1/3}.$$

In figure 2, the quantity $\Delta(V_c/V) = (V_c/V)_{\text{expt}}-(V_c/V)_{\text{calc}}$ is plotted against T/T_c. The experimental values of V are smoothed values taken from the appropriate polynomial at selected temperature values, while $(V_c/V)_{\text{calc}}$ is the ratio calculated from Guggenheim's equation. The values chosen for T_c and V_c are given in table 3. Undoubtedly, the critical constant which is the less certainly known is V_c, and the arrows in figure 2 show the effect of a 1 per cent change in V_c in typical cases. The points for krypton, xenon, and nitrogen lie on the same curve, which has almost the same slope as the curves through the points for argon and oxygen. Changes in the values adopted for V_c for argon and oxygen of about 0.5 per cent would bring the curves for these five substances into almost exact coincidence. There is therefore

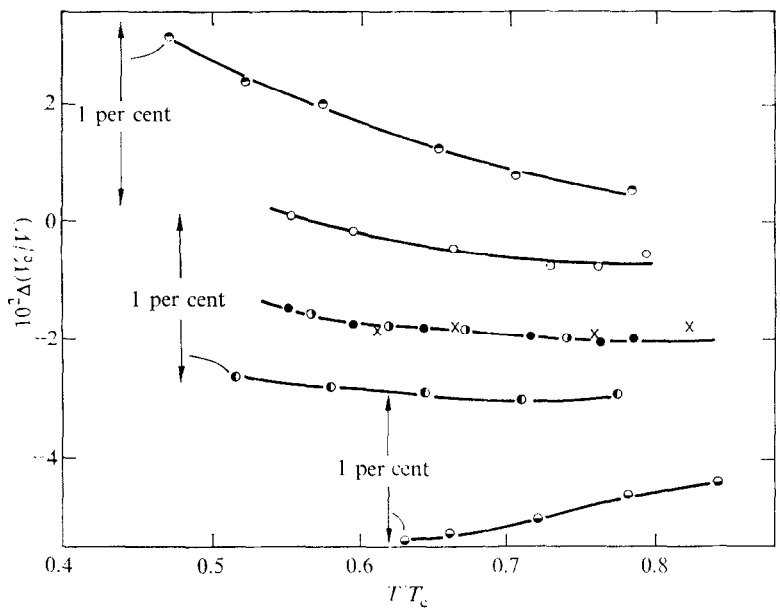


FIGURE 2. Plots of $\Delta(V_c/V)$ against T/T_c, where T denotes temperature measured on the International Practical Scale of 1948, T_c critical temperature, $\Delta(V_c/V) = (V_c/V)_{expt} - (V_c/V)_{calc}$, V denotes molar volume of liquid at temperature T, V_c critical volume, and the subscripts expt and calc refer respectively to experimental volumes and to those calculated from the equation $V_c/V = 1 + \frac{3}{4}(1 - T/T_c) + \frac{1}{4}(1 - T/T_c)^{1/3}$. The arrows show the effect of a 1 per cent change in V_c on the points indicated. ○, Argon; ●, krypton; ◑, xenon; ×, nitrogen; ◐, oxygen; ◓, methane; ◒, carbon monoxide.

TABLE 3. Values of the critical temperature T_c and of the critical volume V_c of the substances investigated. The values of T_c and of V_c(lit.) are regarded as the most reliable values available in the literature. The values of V_c(obs) have been derived from our own results, as described in the text

	T_c/K	V_c(lit)/cm^3 mol^{-1}	V_c(obs)/cm^3 mol^{-1}
Argon	150.6	75.44	75.8
Krypton	209.38	92.24	92.3
Xenon	289.74	118.8	119.8
Oxygen	154.7	74.5	74.9
Nitrogen	126.25	90.09	90.6
Carbon monoxide	132.9	93.02	93.45
Methane	190.78	98.96	100.2
Carbon tetrafluoride	228	146.7	142.4

nothing obviously abnormal in the volume of liquid oxygen over the temperature range studied which might be attributed to dimerization in the liquid. On the other hand, the deviation curve for methane in figure 2 appears to be definitely steeper, and a change of about 1 per cent in V_c for methane which aligned the curve with that for, say, krypton, xenon, and nitrogen at the highest temperature would still leave a discrepancy at lower T/T_c values. This is perhaps evidence of slight quantum effects in methane, possibly related to the rotational movement of the molecules. The curve for carbon monoxide has a slope of a different sign from all other curves, and the difference between this substance on the one hand and the other two diatomic substances on the other may conceivably be due to a greater restraint on the rotation of carbon monoxide molecules in the liquid than on the molecules in liquid nitrogen and oxygen. (It seems certain that in the respective solids carbon monoxide molecules have the most difficulty in re-orientating themselves, perhaps because of the three molecules carbon monoxide has the largest quadrupole moment.)[12]

We shall shortly refer to the values of V_c which can be derived from our own results. If the value so obtained for V_c for carbon monoxide is used in plotting figure 2, the resulting deviation plot lies still lower. On the other hand, use of the only value we can find for V_c for carbon tetrafluoride[13] gives relatively large values of $\Delta(V_c/V)$, between -12×10^{-2} and -18×10^{-2}, and these are approximately halved if the value of V_c given by our own results is used. The slope of the deviation curve for this substance appears to be positive, as for carbon monoxide, but since the deviation plot depends on T_c as well as V_c, and since the experimental error in T_c for carbon tetrafluoride may be greater than that for the other substances, it is impossible to say how closely or otherwise carbon tetrafluoride conforms to the law of corresponding states.

Our measurements make it possible to test the so-called law of the rectilinear diameter over about half the liquid range. At the highest vapour pressures, about 10 bar, reached in our measurements, the gas density ρ_g is still less than 10 per cent of that of the liquid ρ_l, so values of ρ_g can be calculated with adequate accuracy neglecting the third virial coefficient and using the low-temperature values now available for the second virial coefficient B for a number of substances. (When the second virial coefficients were lacking, they could be estimated with sufficient accuracy from corresponding states plots of B/V_c against T/T_c, with the possible exception of carbon tetrafluoride.) The resulting values $\rho_{1/2}$, $=(\rho_l+\rho_g)/2$, should have a precision of a few parts in 10^4. For each substance, $\rho_{1/2}$ was assumed to be a linear function of T of the form

$$\rho_{1/2} = a + b(T_c - T).$$

The densities were expressed for each substance as reciprocal molar volumes, so that $a = 1/V_c$. Using the values of T_c given in table 3, values of a and b were found which minimized $\Sigma\delta^2$, where $\delta = \rho_{1/2}(\text{expt}) - \rho_{1/2}(\text{calc})$. These values of a and b are given in table 4, together with the deviations δ. These are remarkably small, and except for carbon tetrafluoride seldom exceed 0.1 per cent of $\rho_{1/2}$. Nevertheless, for every substance over the range studied, the sign of the deviations follows the same pattern, being negative in the middle of the range and positive at lower and higher

TABLE 4. Values of $\rho_{1/2}$ at regular temperature intervals, where $\rho_{1/2} = \frac{1}{2}(\rho_l + \rho_g)$, ρ_l and ρ_g being the densities of co-existing liquid and vapour phases expressed as reciprocals of the molar volumes. a and b are the values of the coefficients in the equation: $\rho_{1/2}/\mathrm{cm}^{-3}\ \mathrm{mol} = a + b(T_c - T)/\mathrm{K}$, and $\delta = \rho_{1/2}(\mathrm{expt}) - \rho_{1/2}(\mathrm{calc})$

	T/K	$10^2\rho_{1/2}$ / cm^{-3} mol	$10^5\delta$ / cm^{-3} mol	T/K	$10^2\rho_{1/2}$ / cm^{-3} mol	$10^5\delta$ / cm^{-3} mol
Argon	83.82	1.772 0	0.9	105	1.626 8	−1.1
$a = 1.319\,4 \times 10^{-2}$	85	1.764 1	0.9	110	1.593 7	−0.3
$b = 6.764 \times 10^{-5}$	90	1.729 5	0.1	115	1.560 8	0.5
	95	1.694 7	−0.9	120	1.527 7	1.2
	100	1.660 4	−1.3			
Krypton	115.8	1.460 5	0.6	140	1.361 2	−1.5
$a = 1.083\,8 \times 10^{-2}$	117.5	1.453 8	0.7	145	1.341 5	−1.1
$b = 4.019 \times 10^{-5}$	120	1.443 7	0.7	150	1.321 8	−0.6
	125	1.423 2	0.2	155	1.302 3	0
	130	1.402 4	−0.4	160	1.283 0	0.7
	135	1.381 8	−0.9	165	1.263 7	1.5
Xenon	165	1.123 1	0.1	195	1.053 5	−0.2
$a = 8.348\,9 \times 10^{-3}$	170	1.111 5	0.1	200	1.042 1	0
$b = 2.310 \times 10^{-5}$	175	1.100 2	0.2	205	1.030 3	−0.3
	180	1.088 4	0	210	1.018 7	−0.4
	185	1.076 8	0	215	1.007 5	0
	190	1.065 0	−0.2	220	0.996 8	0.8
Oxygen	80	1.855 7	0.6	105	1.680 6	−0.7
$a = 1.335\,9 \times 10^{-2}$	85	1.821 0	0.7	110	1.646 1	−0.4
$b = 6.950 \times 10^{-5}$	90	1.785 3	−0.3	115	1.612 0	0.2
	95	1.750 4	−0.4	120	1.578 2	1.2
	100	1.715 3	−0.7			
Nitrogen	77.355	1.442 8	1.5	92	1.338 9	−1.1
$a = 1.103\,3 \times 10^{-2}$	80	1.423 4	0.4	96	1.311 3	−1.1
$b = 6.912 \times 10^{-5}$	84	1.394 8	−0.5	100	1.284 4	−0.3
	88	1.366 7	−1.0	104	1.259 2	2.1
Carbon monoxide	77.355	1.446 0	1.9	96	1.317 9	−0.7
$a = 1.070\,0 \times 10^{-2}$	80	1.426 7	0.4	100	1.291 1	−0.5
$b = 6.733\,5 \times 10^{-5}$	84	1.398 7	−0.7	104	1.264 3	−0.4
	88	1.371 5	−1.0	108	1.238 1	0.3
	92	1.344 7	−0.8	112	1.212 3	1.5
Methane	90	1.405 2	−0.4	125	1.263 4	−0.7
$a = 9.980\,2 \times 10^{-3}$	95	1.386 1	0.7	130	1.243 3	−0.5
$b = 4.045 \times 10^{-5}$	100	1.366 0	0.8	135	1.223 4	−0.3
	105	1.345 4	0.4	140	1.203 4	0
	110	1.324 7	0	145	1.183 6	0.4
	115	1.304 0	−0.5	150	1.163 8	0.8
	120	1.283 6	−0.7			
Carbon tetrafluoride	95	1.050 5	2.2	145	0.915 6	−2.6
$a = 7.022\,3 \times 10^{-3}$	100	1.037 0	1.7	150	0.902 2	−3.1
$b = 2.603 \times 10^{-5}$	105	1.023 7	1.4	155	0.889 3	−3.0
	110	1.010 5	1.1	160	0.876 3	−2.9
	115	0.997 1	0.8	165	0.864 2	−2.0
	120	0.983 7	0.4	170	0.852 0	−1.2
	125	0.970 2	−0.1	175	0.840 8	0.6
	130	0.956 5	−0.7	180	0.830 6	3.5
	135	0.942 9	−1.4	185	0.821 5	7.4
	140	0.929 3	−2.0			

temperatures. In other words, the rectilinear diameter for all the substances studied in about the lower half of the liquid range is in fact a gentle curve.

If the rectilinear diameter is plotted as the reduced density $\rho_{1/2}/\rho_c$, where ρ_c is the critical density, against the reduced temperature T/T_c, the slope of the line will be $(b/a)T_c$. It was known that this quantity tends to be larger for non-spherical molecules than for argon.[14] The values of $(b/a)T_c$ for the substances we have studied, given by the values of a and b in table 4 and those of T_c in table 3, are as follows: Ar, 0.772; Kr, 0.776; Xe, 0.802; N_2, 0.791; O_2, 0.805; CO, 0.836; CH_4, 0.773; CF_4, 0.845.

The values of V_c given by the values of the parameters a of table 4 are shown in table 3. Since they agree within 1 per cent or so with estimates of V_c based on experiments taken much nearer the critical point, it seems that in spite of the slight curvature in the lower part of the rectilinear diameter, a least-squares line through a plot of $\rho_{1/2}$ against T in this region can give a useful approximation to V_c.

Finally, a relation of the following form has often been used to represent the difference between ρ_l and ρ_g as a function of temperature:

$$\tfrac{1}{2}(\rho_l-\rho_g) = d(T_c-T)^{1/3}.$$

We have used our results to find the best values of c in the more general relation

$$\tfrac{1}{2}(\rho_l-\rho_g) = d(T_c-T)^c,$$

with the following results: Ar, 0.319 7; Kr, 0.318 4; Xe, 0.312 0; N_2, 0.322 2; O_2, 0.312 4; CO, 0.318 4; CH_4, 0.308 5; CF_4, 0.320 0. Thus, in all cases, c is less than $\frac{1}{3}$, the mean value being 0.316 5.

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